

MEDICAID ANALYSIS PROJECT

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-- STEP 1: SCHEMA & TABLE SETUP

```
CREATE SCHEMA IF NOT EXISTS raw;
```

```
CREATE SCHEMA IF NOT EXISTS staging;
```

```
CREATE SCHEMA IF NOT EXISTS analytics;
```

-- STEP 2A: CREATE RAW TABLE: raw.medicaid_drug_utilization

```
CREATE TABLE raw.medicaid_drug_utilization (
```

```
    utilization_type TEXT,
```

```
    state TEXT,
```

```
    ndc TEXT,
```

```
    labeler_code TEXT,
```

```
    product_code TEXT,
```

```
    package_size TEXT,
```

```
    year INT,
```

```
    quarter INT,
```

```
    suppression_used BOOLEAN,
```

```
    product_name TEXT,
```

```
    units_reimbursed NUMERIC,
```

```
    number_of_prescriptions NUMERIC,
```

```
    total_amount_reimbursed NUMERIC,
```

```
    medicaid_amount_reimbursed NUMERIC,
```

```
    non_medicaid_amount_reimbursed NUMERIC
```

```
);
```

-- STEP 2B: VERIFY RAW TABLE EXISTS

```
SELECT *  
FROM raw.medicaid_drug_utilization  
LIMIT 5;
```

-- STEP 3: IMPORT U.S MEDICAID DATA (FROM 2016 TO 2025)

-- STEP 3A: IMPORT 2016 DATA

```
COPY raw.medicaid_drug_utilization  
FROM 'F:\Project-2\State Drug Utilization Data - 2016.csv'  
DELIMITER ','  
CSV HEADER;
```

-- STEP 3B: IMPORT 2017 DATA

```
COPY raw.medicaid_drug_utilization  
FROM 'F:\Project-2\State Drug Utilization Data - 2017.csv'  
DELIMITER ','  
CSV HEADER;
```

-- STEP 3C: IMPORT 2018 DATA

```
COPY raw.medicaid_drug_utilization  
FROM 'F:\Project-2\State Drug Utilization Data - 2018.csv'  
DELIMITER ','  
CSV HEADER;
```

-- STEP 3D: IMPORT 2019 DATA

```
COPY raw.medicaid_drug_utilization
```

```
FROM 'F:\Project-2\State Drug Utilization Data - 2019.csv'  
DELIMITER ','  
CSV HEADER;
```

-- STEP 3E: IMPORT 2020 DATA

```
COPY raw.medicaid_drug_utilization  
FROM 'F:\Project-2\State Drug Utilization Data - 2020.csv'  
DELIMITER ','  
CSV HEADER;
```

-- STEP 3F: IMPORT 2021 DATA

```
COPY raw.medicaid_drug_utilization  
FROM 'F:\Project-2\State Drug Utilization Data - 2021.csv'  
DELIMITER ','  
CSV HEADER;
```

-- STEP 3G: IMPORT 2022 DATA

```
COPY raw.medicaid_drug_utilization  
FROM 'F:\Project-2\State Drug Utilization Data - 2022.csv'  
DELIMITER ','  
CSV HEADER;
```

-- STEP 3H: IMPORT 2023 DATA

```
COPY raw.medicaid_drug_utilization  
FROM 'F:\Project-2\State Drug Utilization Data - 2023.csv'  
DELIMITER ','  
CSV HEADER;
```

-- STEP 3I: IMPORT 2024 DATA

```
COPY raw.medicaid_drug_utilization
FROM 'F:\Project-2\State Drug Utilization Data - 2024.csv'
DELIMITER ','
CSV HEADER;
```

-- STEP 3J: IMPORT 2025 DATA

```
COPY raw.medicaid_drug_utilization
FROM 'F:\Project-2\State Drug Utilization Data - 2025.csv'
DELIMITER ','
CSV HEADER;
```

-- STEP 4: VERIFY IMPORTED FILES FROM 2016 TO 2025

-- STEP 4A: VERIFY EACH YEAR IS PRESENT

```
SELECT DISTINCT year
FROM raw.medicaid_drug_utilization
ORDER BY year;
```

-- STEP 4B: COUNT ROWS PER YEAR

```
SELECT
    year,
    COUNT(*) AS total_rows
FROM raw.medicaid_drug_utilization
GROUP BY year
ORDER BY year;
```

-- STEP 4C: CHECK SUPPRESSED VS NON-SUPPRESSED BY YEAR

```
SELECT
    year,
    COUNT (*) AS total_rows,
    COUNT (*) FILTER (WHERE suppression_used = TRUE) AS suppressed_rows,
    COUNT (*) FILTER (WHERE suppression_used = FALSE) AS
non_suppressed_rows
FROM raw.medicaid_drug_utilization
GROUP BY year
ORDER BY year;
```

-- STEP 4D: QUICK CHECK OF YEAR-QUARTER COMBINATIONS

```
SELECT
    year,
    quarter,
    COUNT (*) AS row_count
FROM raw.medicaid_drug_utilization
GROUP BY year, quarter
ORDER BY year, quarter;
```

-- STEP 4E: CHECK NULL BEHAVIOR BY YEAR

```
SELECT
    year,
    COUNT (units_reimbursed) AS units_non_null,
    COUNT (number_of_prescriptions) AS prescriptions_non_null,
    COUNT (total_amount_reimbursed) AS amount_non_null
FROM raw.medicaid_drug_utilization
```

```
GROUP BY year  
ORDER BY year;
```

-- STEP 4F: TOTAL DATASET SIZE

```
SELECT COUNT (*)  
FROM raw.medicaid_drug_utilization;
```

-- STEP 5: CREATE CLEAN STAGING VIEW

```
CREATE VIEW staging.medicaid_clean AS  
SELECT  
    utilization_type,  
    state,  
    ndc,  
    labeler_code,  
    product_code,  
    package_size,  
    year,  
    quarter,  
    product_name,  
    suppression_used,  
  
    -- Clean numeric fields (ignore suppressed rows)  
    CASE  
        WHEN suppression_used = FALSE THEN units_reimbursed  
        ELSE NULL  
    END AS units_reimbursed,
```

```
CASE
    WHEN suppression_used = FALSE THEN number_of_prescriptions
    ELSE NULL
END AS number_of_prescriptions,
```

```
CASE
    WHEN suppression_used = FALSE THEN total_amount_reimbursed
    ELSE NULL
END AS total_amount_reimbursed,
```

```
CASE
    WHEN suppression_used = FALSE THEN medicaid_amount_reimbursed
    ELSE NULL
END AS medicaid_amount_reimbursed,
```

```
CASE
    WHEN suppression_used = FALSE THEN non_medicaid_amount_reimbursed
    ELSE NULL
END AS non_medicaid_amount_reimbursed
```

```
FROM raw.medicaid_drug_utilization;
```

```
-- STEP 6: VALIDATE VIEW
```

```
-- STEP 6A: VALIDATE VIEW'S TOTAL DATASET SIZE
```

```
SELECT COUNT(*)
```

```
FROM staging.medicaid_clean;
```

-- STEP 6B: CHECK TOTAL ROWS & NON_NULL VALUES FROM VIEW

SELECT

 COUNT(*) AS total_rows,

 COUNT(units_reimbursed) AS units_non_null,

 COUNT(number_of_prescriptions) AS prescriptions_non_null,

 COUNT(total_amount_reimbursed) AS amount_non_null

FROM staging.medicaid_clean;

-- STEP 7: FEATURE ENGINEERING

CREATE VIEW staging.medicaid_enriched AS

SELECT

 *,

 -- Cost per prescription

 CASE

 WHEN number_of_prescriptions > 0

 THEN total_amount_reimbursed / number_of_prescriptions

 ELSE NULL

 END AS cost_per_prescription,

 -- Cost per unit

 CASE

 WHEN units_reimbursed > 0

 THEN total_amount_reimbursed / units_reimbursed

 ELSE NULL

```

END AS cost_per_unit,

-- Units per prescription (usage intensity)
CASE
    WHEN number_of_prescriptions > 0
    THEN units_reimbursed / number_of_prescriptions
    ELSE NULL
END AS units_per_prescription,

-- Medicaid dependency ratio
CASE
    WHEN total_amount_reimbursed > 0
    THEN medicaid_amount_reimbursed / total_amount_reimbursed
    ELSE NULL
END AS medicaid_dependency_ratio

FROM staging.medicaid_clean;

-- STEP 8: VALIDATE FEATURES
SELECT *
FROM staging.medicaid_enriched
LIMIT 10;

-- STEP 9: KPI / AGGREGATION LAYER
-- STEP 9A1: YEAR-LEVEL KPI VIEW

CREATE VIEW analytics.kpi_yearly AS

```

```

SELECT
    year,
    COUNT(*) AS row_count,
    COUNT(*) FILTER (WHERE suppression_used = TRUE) AS suppressed_rows,
    COUNT(*) FILTER (WHERE suppression_used = FALSE) AS
non_suppressed_rows,

    SUM(total_amount_reimbursed) AS total_spend,
    SUM(medicaid_amount_reimbursed) AS medicaid_spend,
    SUM(non_medicaid_amount_reimbursed) AS non_medicaid_spend,
    SUM(number_of_prescriptions) AS total_prescriptions,
    SUM(units_reimbursed) AS total_units,

    AVG(cost_per_prescription) AS avg_cost_per_prescription,
    AVG(cost_per_unit) AS avg_cost_per_unit,
    AVG(units_per_prescription) AS avg_units_per_prescription,
    AVG(medicaid_dependency_ratio) AS avg_medicaid_dependency_ratio

FROM staging.medicaid_enriched
GROUP BY year
ORDER BY year;

```

-- STEP 9A2: VALIDATION OF YEAR-LEVEL KPI VIEW

```

SELECT *
FROM analytics.kpi_yearly;

```

-- STEP 9B1: STATE-LEVEL KPI VIEW

```

CREATE VIEW analytics.kpi_state AS
SELECT
    state,
    COUNT(*) AS row_count,
    COUNT(*) FILTER (WHERE suppression_used = TRUE) AS suppressed_rows,
    COUNT(*) FILTER (WHERE suppression_used = FALSE) AS
non_suppressed_rows,

    SUM(total_amount_reimbursed) AS total_spend,
    SUM(medicaid_amount_reimbursed) AS medicaid_spend,
    SUM(non_medicaid_amount_reimbursed) AS non_medicaid_spend,
    SUM(number_of_prescriptions) AS total_prescriptions,
    SUM(units_reimbursed) AS total_units,

    AVG(cost_per_prescription) AS avg_cost_per_prescription,
    AVG(cost_per_unit) AS avg_cost_per_unit,
    AVG(units_per_prescription) AS avg_units_per_prescription,
    AVG(medicaid_dependency_ratio) AS avg_medicaid_dependency_ratio

FROM staging.medicaid_enriched
GROUP BY state
ORDER BY total_spend DESC;

-- STEP 9B2: VALIDATION OF STATE-LEVEL KPI VIEW
SELECT *
FROM analytics.kpi_state;

```

-- STEP 9C1: STATE x YEAR KPI VIEW

CREATE VIEW analytics.kpi_state_year AS

SELECT

state,

year,

COUNT(*) AS row_count,

COUNT(*) FILTER (WHERE suppression_used = TRUE) AS suppressed_rows,

COUNT(*) FILTER (WHERE suppression_used = FALSE) AS
non_suppressed_rows,

SUM(total_amount_reimbursed) AS total_spend,

SUM(medicaid_amount_reimbursed) AS medicaid_spend,

SUM(non_medicaid_amount_reimbursed) AS non_medicaid_spend,

SUM(number_of_prescriptions) AS total_prescriptions,

SUM(units_reimbursed) AS total_units,

AVG(cost_per_prescription) AS avg_cost_per_prescription,

AVG(cost_per_unit) AS avg_cost_per_unit,

AVG(units_per_prescription) AS avg_units_per_prescription,

AVG(medicaid_dependency_ratio) AS avg_medicad_dependency_ratio

FROM staging.medicad_enriched

GROUP BY state, year

ORDER BY state, year;

-- STEP 9C2: VALIDATION OF STATE × YEAR KPI VIEW

```
SELECT *  
FROM analytics.kpi_state_year  
LIMIT 20;
```

-- STEP 9C3: COUNT TOTAL ROWS

```
SELECT  
    COUNT (*) AS total_ROWS  
FROM analytics.kpi_state_year;
```

-- STEP 9D1: DRUG-LEVEL KPI VIEW

```
CREATE OR REPLACE VIEW analytics.kpi_drug AS  
SELECT  
    product_name,  
    COUNT(*) AS row_count,  
    COUNT(*) FILTER (WHERE suppression_used = TRUE) AS suppressed_rows,  
    COUNT(*) FILTER (WHERE suppression_used = FALSE) AS  
non_suppressed_rows,  
  
    SUM(total_amount_reimbursed) AS total_spend,  
    SUM(number_of_prescriptions) AS total_prescriptions,  
    SUM(units_reimbursed) AS total_units,  
  
    AVG(cost_per_prescription) AS avg_cost_per_prescription,  
    AVG(cost_per_unit) AS avg_cost_per_unit,
```

```
    AVG(units_per_prescription) AS avg_units_per_prescription,  
    AVG(medicaid_dependency_ratio) AS avg_medicaid_dependency_ratio
```

```
FROM staging.medicaid_enriched  
WHERE suppression_used = FALSE  
GROUP BY product_name  
ORDER BY total_spend DESC;
```

-- STEP 9D2: VALIDATION OF DRUG LEVEL KPI VIEW

```
SELECT *  
FROM analytics.kpi_drug  
LIMIT 20;
```

-- STEP 9D3: VERIFY TOTAL_DRUGS & NON_NULL_SPEND FROM DRUG LEVEL KPI VIEW

```
SELECT  
    COUNT(*) AS total_drugs,  
    COUNT(total_spend) AS non_null_spend  
FROM analytics.kpi_drug;
```

-- STEP 9D4: VERIFY REAL_DRUG_NAMES & ACTUAL_NUMBERS (NOT_NULL) FROM DRUG LEVEL KPI VIEW

```
SELECT *  
FROM analytics.kpi_drug  
ORDER BY total_spend DESC  
LIMIT 5;
```

-- STEP 9E1: DRUG SHARE (PARETO READY)

CREATE VIEW analytics.kpi_drug_share AS

SELECT

*,

-- Spend share (% of total spend)

total_spend / SUM(total_spend) OVER () AS spend_share,

-- Prescription share (% of total prescriptions)

total_prescriptions / SUM(total_prescriptions) OVER () AS
prescription_share

FROM analytics.kpi_drug;

-- STEP 9E2: VALIDATION OF DRUG SHARE (PARETO CHART)

SELECT *

FROM analytics.kpi_drug_share

ORDER BY total_spend DESC

LIMIT 10;

-- STEP 9F1: PARETO ANALYSIS (CUMULATIVE %)

CREATE VIEW analytics.kpi_drug_pareto AS

SELECT

*,

-- Cumulative spend %

```
SUM(spend_share) OVER (  
    ORDER BY total_spend DESC  
    ) AS cumulative_spend_share  
  
FROM analytics.kpi_drug_share;
```

-- STEP 9F2: VALIDATION OF PARETO ANALYSIS

```
SELECT  
    product_name,  
    total_spend,  
    spend_share,  
    cumulative_spend_share  
FROM analytics.kpi_drug_pareto  
ORDER BY total_spend DESC  
LIMIT 20;
```

-- STEP 9F3: DRUGS CONTRIBUTING TO 80% OF TOTAL SPEND

```
SELECT *  
FROM analytics.kpi_drug_pareto  
WHERE cumulative_spend_share <= 0.80;
```

-- STEP 9G1: FFSU vs MCOU KPI VIEW

```
CREATE VIEW analytics.kpi_utilization_type AS  
SELECT  
    utilization_type,
```

```
    COUNT(*) AS row_count,  
    COUNT(*) FILTER (WHERE suppression_used = TRUE) AS suppressed_rows,  
    COUNT(*) FILTER (WHERE suppression_used = FALSE) AS  
non_suppressed_rows,
```

```
    SUM(total_amount_reimbursed) AS total_spend,  
    SUM(medicaid_amount_reimbursed) AS medicaid_spend,  
    SUM(non_medicaid_amount_reimbursed) AS non_medicaid_spend,  
    SUM(number_of_prescriptions) AS total_prescriptions,  
    SUM(units_reimbursed) AS total_units,
```

```
    AVG(cost_per_prescription) AS avg_cost_per_prescription,  
    AVG(cost_per_unit) AS avg_cost_per_unit,  
    AVG(units_per_prescription) AS avg_units_per_prescription,  
    AVG(medicaid_dependency_ratio) AS avg_medicaid_dependency_ratio
```

```
FROM staging.medicaid_enriched  
GROUP BY utilization_type  
ORDER BY total_spend DESC;
```

-- STEP 9G2: VALIDATION OF FFSU VS MCOU KPI

```
SELECT *  
FROM analytics.kpi_utilization_type;
```

-- STEP 10: ANALYSIS FOR SUPPRESSED DRUGS

-- STEP 10A1: SUPPRESSION BY YEAR

```

CREATE VIEW analytics.kpi_suppression_year AS
SELECT
    year,
    COUNT(*) AS total_rows,
    COUNT(*) FILTER (WHERE suppression_used = TRUE) AS suppressed_rows,
    COUNT(*) FILTER (WHERE suppression_used = FALSE) AS
non_suppressed_rows,
    COUNT(*) FILTER (WHERE suppression_used = TRUE)::NUMERIC / COUNT(*)
AS suppression_rate
FROM staging.medicaid_clean
GROUP BY year
ORDER BY year;

```

-- STEP 10A2: VALIDATION OF SUPPRESSION BY YEAR

```

SELECT *
FROM analytics.kpi_suppression_year;

```

-- STEP 10B1: SUPPRESSION BY STATE

```

CREATE VIEW analytics.kpi_suppression_state AS
SELECT
    state,
    COUNT(*) AS total_rows,
    COUNT(*) FILTER (WHERE suppression_used = TRUE) AS suppressed_rows,
    COUNT(*) FILTER (WHERE suppression_used = FALSE) AS
non_suppressed_rows,
    COUNT(*) FILTER (WHERE suppression_used = TRUE)::NUMERIC / COUNT(*)
AS suppression_rate

```

```
FROM staging.medicaid_clean
GROUP BY state
ORDER BY suppressed_rows DESC;
```

-- STEP 10B2: VALIDATION OF SUPPRESSION BY STATE

```
SELECT *
FROM analytics.kpi_suppression_state
LIMIT 20;
```

-- STEP 10C1: SUPPRESSION BY UTILIZATION TYPE

```
CREATE VIEW analytics.kpi_suppression_utilization_type AS
SELECT
    utilization_type,
    COUNT(*) AS total_rows,
    COUNT(*) FILTER (WHERE suppression_used = TRUE) AS suppressed_rows,
    COUNT(*) FILTER (WHERE suppression_used = FALSE) AS
non_suppressed_rows,
    COUNT(*) FILTER (WHERE suppression_used = TRUE)::NUMERIC / COUNT(*)
AS suppression_rate
FROM staging.medicaid_clean
GROUP BY utilization_type
ORDER BY suppressed_rows DESC;
```

-- STEP 10C2: VALIDATION OF SUPPRESSION BY UTILIZATION TYPE

```
SELECT *
FROM analytics.kpi_suppression_utilization_type;
```

-- STEP 10D1: MOST FREQUENTLY SUPPRESSED DRUGS

```
CREATE VIEW analytics.kpi_suppressed_drug AS
SELECT
    product_name,
    COUNT(*) AS suppressed_count
FROM staging.medicaid_clean
WHERE suppression_used = TRUE
GROUP BY product_name
ORDER BY suppressed_count DESC;
```

-- STEP 10D2: MOST FREQUENTLY SUPPRESSED DRUGS

```
SELECT *
FROM analytics.kpi_suppressed_drug
LIMIT 20;
```